

# Shrabya Bhattarai

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## EDUCATION

### University of Southern Mississippi

Bachelor of Science in Computer Engineering

Hattiesburg, MS

Aug 2024 – May 2028

- GPA: 3.91/4.0 | Relevant Coursework: Digital Electronics, Computer Networks, Operating Systems, Embedded Systems, Data Structures

## TECHNICAL SKILLS

**Languages:** C, Python, JavaScript, Bash, SQL, Assembly (ARM)

**Data & Analytics:** Pandas, NumPy, SQL querying, data cleaning, CRM databases, automated reporting

**Embedded & Systems:** STM32, Raspberry Pi, UART/I2C/SPI, POSIX API, GNU Make, GDB, Valgrind

**Tools & Frameworks:** Git, Linux, Flask, OpenCV, TensorFlow Lite, pytest, Makefile, VS Code

## EXPERIENCE

### 4th Dimension Marketing Solutions

Cincinnati, OH

Data Analyst Intern

May 2026 – Aug 2026

- Write Python scripts that automate the collection and organization of daily field sales data into structured databases, replacing manual entry across multiple sales locations.
- Clean and validate raw sales records from physical marketing campaigns, resolving inconsistencies before the data enters reporting and analysis workflows.
- Build automated reporting pipelines that generate weekly sales performance summaries for the marketing team, reducing turnaround on reporting requests.
- Analyze multi-location sales data to identify performance trends and surface insights used to adjust campaign strategy.
- Maintain a centralized CRM database using SQL to track leads, sales figures, and campaign outcomes across the team.

### TechMatrix

Kathmandu, Nepal

Junior Software Engineer Intern

May 2023 – Aug 2024

- Built and maintained RESTful APIs in Python (Flask) to support a client-facing logistics dashboard, focusing on performance and reliability.
- Collaborated with a team of engineers using Git and Agile/Scrum methodology, participating in daily standups, sprint planning, and code reviews.
- Wrote unit and integration tests using pytest to ensure code quality and catch bugs before production deployment.

### Sugam Engineering Pvt. Ltd.

Kathmandu, Nepal

Embedded Systems Developer Intern

Dec 2021 – Mar 2022

- Developed firmware in C for an STM32-based temperature and humidity monitoring device, handling sensor interfacing and real-time data logging.
- Worked on optimizing event handling routines to improve performance and reduce unnecessary CPU load on a resource-constrained microcontroller.
- Documented hardware schematics and contributed to internal technical documentation used for onboarding new team members.

## PROJECTS

### WildfireWatch — Onboard AI for UAV Detection

Python, OpenCV, TensorFlow Lite, C

Research Project | University of Southern Mississippi

Jan 2026 – Present

- Conducting a literature survey on onboard AI for UAV systems, reviewing papers across embodied navigation, multi-agent coordination, and edge inference techniques.
- Synthesizing findings across 35+ papers to support a research publication on autonomous wildfire detection using AI-equipped UAV platforms.
- Exploring lightweight model deployment strategies for resource-constrained onboard hardware under faculty supervision.

### SmartSense — Raspberry Pi Environmental Monitor

Python, C, Raspberry Pi, SQLite

Personal Project

Aug 2025 – Nov 2025

- Built a real-time environmental monitoring system on Raspberry Pi using Python, interfacing a DHT22 temperature/humidity sensor over GPIO and logging data to a local SQLite database.
- Wrote a lightweight C module to handle low-level GPIO polling and pass structured sensor readings to the Python application layer via subprocess communication.
- Developed a terminal dashboard using Python's curses library to display live sensor readings and historical trends without requiring an external display or GUI.

## ACTIVITIES & LEADERSHIP

### USM Open Source Community (OSC)

University of Southern Mississippi

Social Media Lead & Co-Founder

March 2026 – Present

- Co-founded the OSC club and manage the @oscusm social media presence, growing the community to 200+ student members through workshops, open-source contribution drives, and event coordination.